



Dynamic traffic light sequencing (Credit: www.mdpi.com)

sequences circumvents or avoids problems that usually arise with systems that use image-processing and beam-interruption techniques. RFID technology with appropriate algorithm and database are applied to a multi-vehicle, multi-lane and multi-road junction area to provide an efficient time-management scheme.

A dynamic time schedule is worked out for the passage of each column. The dynamic sequence algorithm adjusts itself even with the presence of some extreme cases. The system can emulate the judgment of a traffic police officer on duty, by considering the number of vehicles in each column and routing proprieties.

Emergency vehicle notification system. An in-vehicle e-call is generated manually by the vehicle occupant(s), or automatically via activation of in-vehicle sensors after an accident. The e-call carries both voice and data directly to the nearest emergency point (normally a public safety answering point). The voice call enables the vehicle occupant(s) to communicate with the trained e-call operator and simultaneously a minimum set of data—that includes vehicle identification, information about the incident, time of occurrence, precise location and direction in which the vehicle was travelling—is sent to the e-call operator for taking necessary action. In most developed

countries, e-call is a standard option for all approved vehicle types.

Road traffic cooperative system. A cooperative system in road traffic means road operators, infrastructure, vehicles, their drivers and other road users cooperate to deliver the most efficient, safe, secure and comfortable journey. The system is based on communication cooperation on the road.

Data available from vehicles is acquired and transmitted to a server for central fusion and processing. This data can be used to detect events such as rain (wiper activity) and congestion (frequent braking activities). The server processes a driving recommendation dedicated to a single or a specific group of drivers, and transmits it wirelessly to the vehicle(s). The goal of a cooperative system is to use

and plan communication and sensor infrastructure to test road safety.

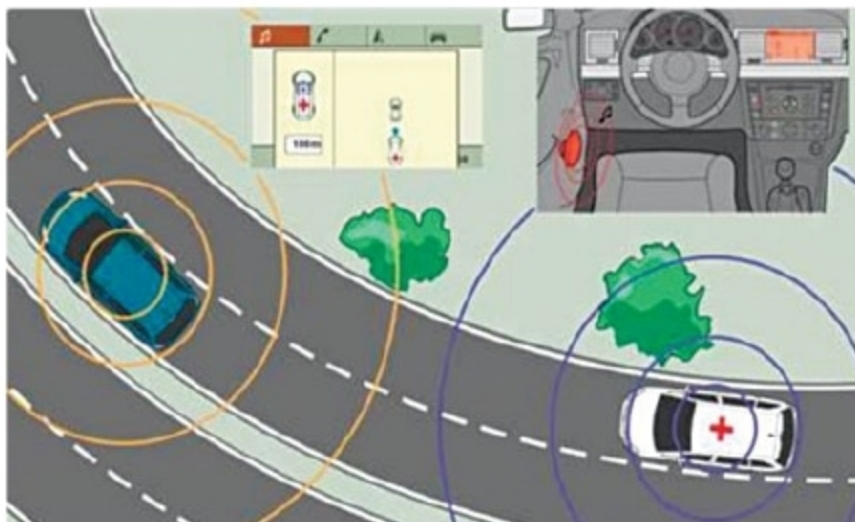
Varying speed limits based on road congestion. Experimentation with variable speed limits that change with road congestion and other factors is being pursued. Typically, such speed limits generally deteriorate during poor conditions, rather than improving in good conditions. One example is Britain's M25 motorway, which circumnavigates London. Variable speed limits combined with automated enforcement result in savings in journey times, smoother-flowing traffic and a fall in the number of accidents.

How a basic ITMS works

A basic ITMS assists road authorities in maximising operational performance and reliability. This assistance is across all aspects of the road network. It is important to manage transport problems by integrating technologies, such as car-navigation system, traffic-signal-control system, variable message signs, automatic number plate recognition or speed cameras, and other systems that integrate live data and feedback from multiple sources, such as parking guidance and information systems, weather information, bridge de-icing systems and the like.

Additionally, predictive techniques are used for advanced modelling and comparison with historical baseline data. Based on real-time and historical data, traffic signal cycles are altered on real-time basis, using in-situ sensors or surveillance devices, to respond to changing traffic conditions.

A simple and basic system comprises a traffic-control centre, a traffic information service centre, a cellular mobile communication system and an in-vehicle terminal (GPS tracer), connected with each other via a wireless communication network (satellites or telecom towers). The



Emergency vehicle notification system wherein an emergency call is generated by the activation of in-vehicle sensors (Credit: www.slideshare.net)